

Interaktion mit maschinellen Agenten



Teil 1: Was sind soziale Roboter?

Teil 2: Soziale Kognitionsforschung mit Robotern

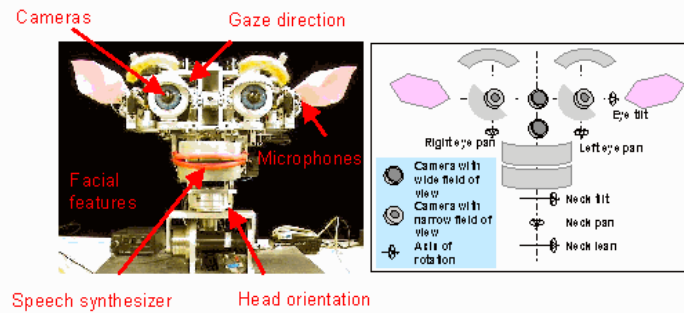
Teil 1: Was sind soziale Roboter?



1. Evolution sozialer Roboter
2. Design sozialer Roboter
3. Anwendung sozialer Roboter



- Der Begriff „**sozialer Roboter**“ wurde erstmals 1935 erwähnt und als abwertender Begriff für eine Person mit einer kalten und distanzierten Persönlichkeit verwendet
- 1961 wurde der erste Roboter „Unimate“ in einer **alltäglichen Umgebung** installiert (General Motors Plant). Es kam schon da die Befürchtung auf, ob der Roboter den Menschen am Arbeitsplatz ersetzen würde
- Erste **wissenschaftliche Erwähnung** eines sozialen Roboters im Jahr 1978: Serviceroboter, die über allgemeine kognitive Fähigkeiten verfügen, sollten auch soziale Fähigkeiten haben
- 1997 wurde **Kismet** als erster soziale Roboter systematisch in der Forschung an den MIT Media Labs eingesetzt

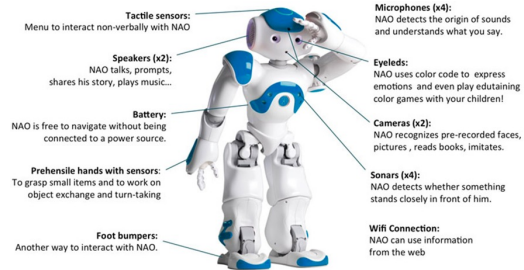


- In den frühen 2000er Jahren haben mehrere Unternehmen soziale Roboter **zum Kauf** angeboten:
 - **Keepon** (2003) by Hideki Kozima
 - **NAO** (2006) by Aldebaran Robotics
 - **Paro** (2006)
 - **Baxter** (2011) by Rethink Robotics



Keepon

<https://www.engadget.com>



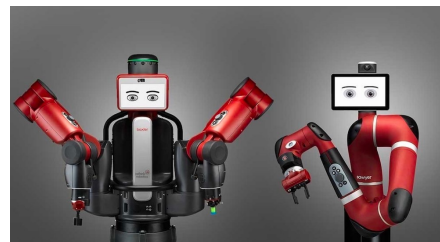
NAO

<https://www.wevolver.com/wevolver.staff/nao.robot>



Paro

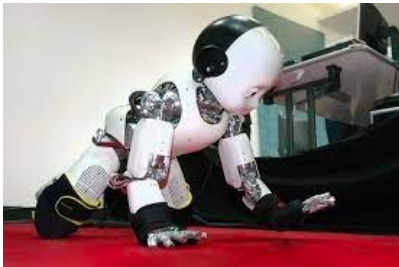
<http://www.icsr2017.org/keynote-shibata.html>



Baxter & Sawyer

<https://www.therobotreport.com/rethink-robotics-shutdown/>

- In den 2000er Jahren wurden viele soziale Roboterplattformen **für die Forschung** entwickelt:
 - **Icub** (2009) für Entwicklungsrobotik
 - **Kaspar** (2009) für Forschung und Therapie bei ASD
 - **Octavia** (2009) für Mensch-Roboter Interaktion mit Erwachsenen
 - **Justin** (2009; by DLR) für Forschung zur Sicherheit in der HRI



Icub

<https://team.inria.fr/larsen/robots/icub/>



Kaspar

<https://www.herts.ac.uk/kaspar/the-social-robot>



Octavia

<https://robots.ieee.org/robots/octavia>



Justin

<https://www.dlr.de/content/en/institutes/institute-of-robotics-and-mechatronics.html>

- Soziale Roboter mit ausgefeilten sozialen Fähigkeiten kann man heute "**von der Stange**" kaufen (für ein paar hundert Euro):
 - Cozmo
 - Maki
 - Buddy
 - Cognitoys
 - Miko
 - und viele mehr ...



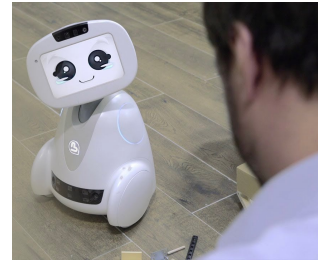
Cozmo

<https://www.n-tv.de/technik>



Maki

<https://www.wevolver.com>



Buddy

<https://www.buddytherobot.com/>



Cognitoys

<https://www.trendhunter.com/>

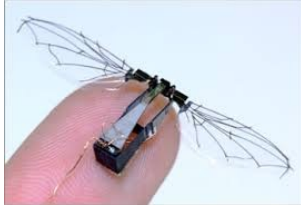


Miko

<https://www.therobotreport.com>



Biologically inspired



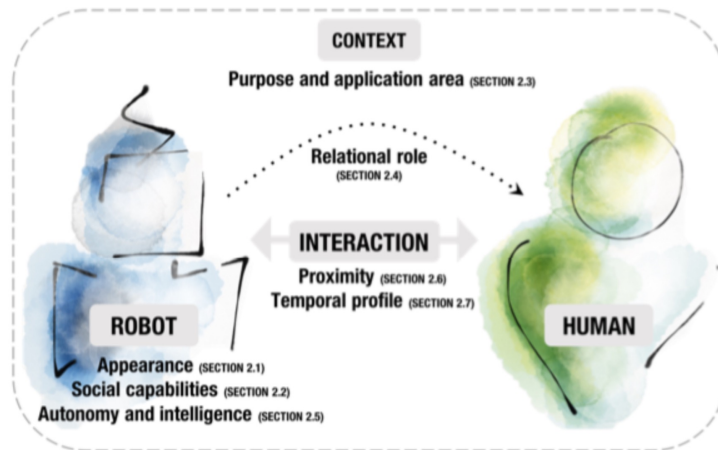
Functionally designed



- Human-oriented perception
- Natural human-robot interaction
- Readable social cues
- Real-time performance

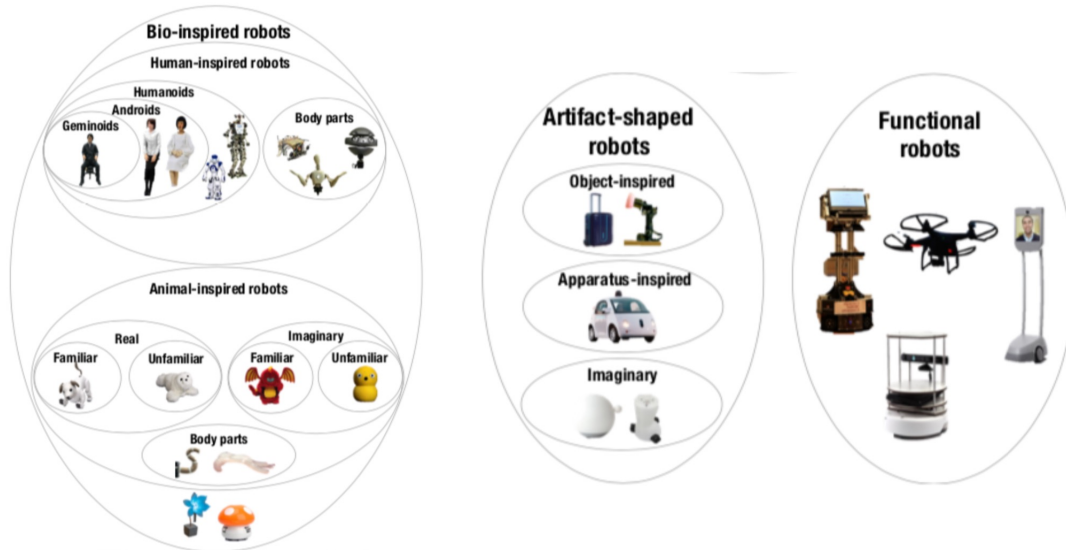
} General issues in robotics

- | | | |
|--|---|------------------------------------|
| <ul style="list-style-type: none">▪ Human-oriented perception▪ Natural human-robot interaction▪ Readable social cues▪ Real-time performance | } | General issues in robotics |
| <ul style="list-style-type: none">▪ Embodiment▪ Emotion▪ Dialogue▪ Personality▪ Human-oriented perception▪ User modeling▪ Social Learning▪ Intentionality | } | Specific issues in social robotics |



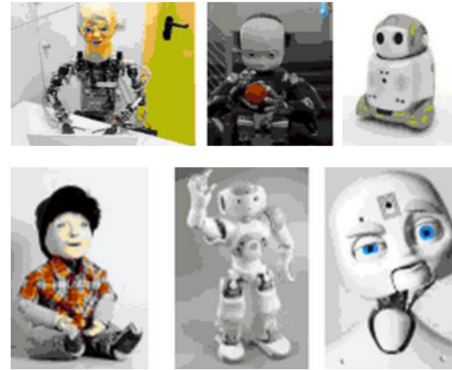


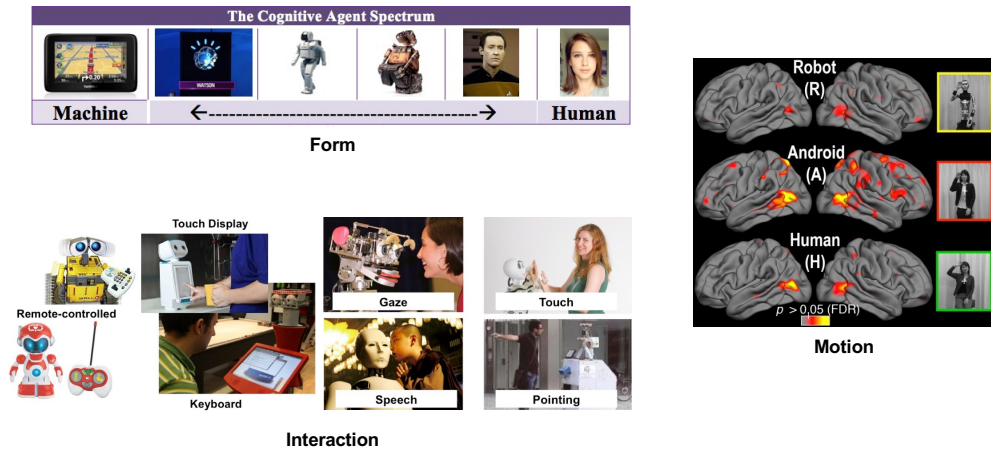
Appearance

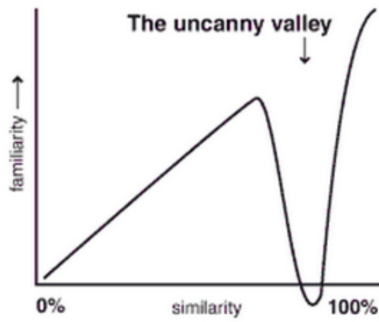


Anthropomorphism

- *anthropos* for human and *morphe* for shape or form
- People's tendency to attribute human characteristics to agents
- Default-mode when understanding and predicting behavior
- Perfect balance between illusion and functionality







How much human-likeness do we want in non-human objects?

- Entertainment and toy robots
- Establishment of human-creature interaction
- Purpose is to create robotic companions
- Avoiding the uncanny valley is easier than in human-humanoid interaction



- Exaggeration of distinctive or striking features
- Simplified or stereotypical representations
- Interaction biases can be used to stimulate certain cognitive mechanisms (e.g. attention)

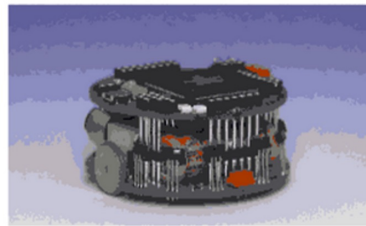


- Embodiment should reflect the task that the robot needs to perform
- Design of physical features is guided by operational objectives
- Design most common for service and toy robots



Embodiment

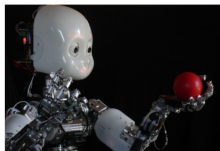
- Establishes a basis for structural coupling by creating the potential for mutual perturbation between system and environment
- The more the robot can perturb an environment, the more embodied it is
- Design of effectors and sensors with which robot can interact with the world



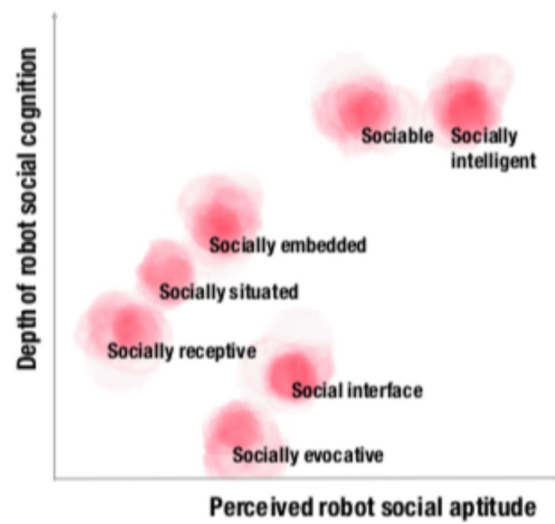


Social Capabilities

1. Communication using natural language
2. Expressing affective behaviors
3. Human centered perception
4. Possessing a distinctive personality or character
5. Modeling social aspects of humans
6. Learning / developing social skills
7. Establishing / maintaining social relationships



Increase of socialness ↓	▪ Socially evocative	rely on tendency to anthropomorphize, evoke emotional acts
	▪ Social interface	natural interface, social cues, communication skills, shallow models of social cognition
	▪ Socially receptive	socially passive but benefit from interaction, more social competencies than social interface robots
	▪ Sociable	Pro-actively engage with humans to satisfy social aims, deep models of social cognition
	▪ Socially situated	surrounded by social environment which they perceive and interact with
	▪ Socially embedded	Socially situated, coupled with environment, aware of human interactional structure
	▪ Socially intelligent	Show aspects of human social intelligence, deep models of social cognition



Verbal and Nonverbal Communication

- Exchange of information between interaction partners
- Modulated by social cues
- Three types:
 - Low-level (only basic language skills)
 - Non-verbal (body posture, gestures, movements)
 - Natural language (person features, situation)



Kismet



Icat



Saya

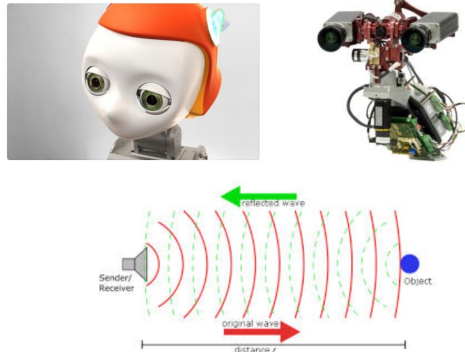


Human Centered Perception

- Passive (computer vision, language recognition. Sensors: cameras)
- Active (localizing humans in dynamic settings. Sensors: ladar, sonar)

Perceptual skills for robots

- People tracking
- Speech recognition
- Gesture recognition
- Facial perception
 - Detection
 - Recognition
 - Expressions
 - Gaze tracking

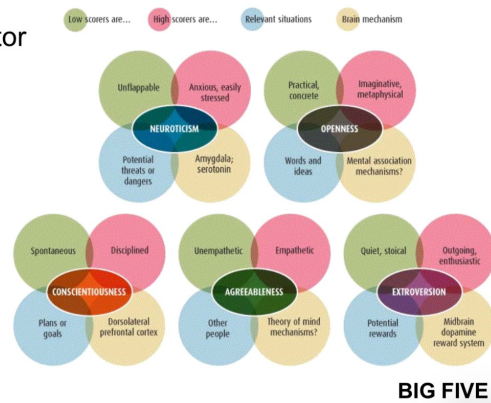


Personality

= Set of distinctive qualities that distinguish individuals from each other

- BIG FIVE, Myers-Briggs Type Indicator

- Personality in social robots
 - Tool-like
 - Pet or creature
 - Cartoon
 - Artificial being
 - Human-like



Models of Social Cognition

- Quantitative: Evaluation of parameters and metrics
- Qualitative: Subjective aspects of user behavior
- Static: Defined a priori
- Dynamic: Adapted or learned

Purpose

- Help robot understand human behavior and dialogue (e.g., language)
- Shape and control feedback given to the user (e.g., motion and pacing)
- Accommodate users with varying skills, experience, and knowledge

Learning / Developing Social Skills

Purpose

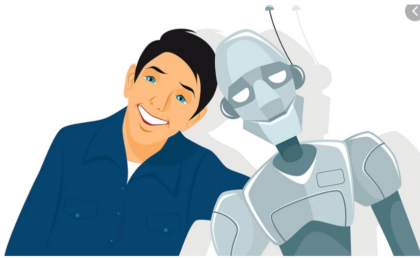
- Transfer skills, tasks, and information
- Match between mental models of interaction partners
- Essential for improving communication, interaction and sharing knowledge



Approaches

- Create sequences of known behavior to match human model
- Match observations to known behaviors
- Animal training techniques
- Conditioning / reinforcement learning

Establishing Social Relationships





Autonomy

Extent to which a robot can operate on tasks it was designed (or that it creates for itself) for without external intervention

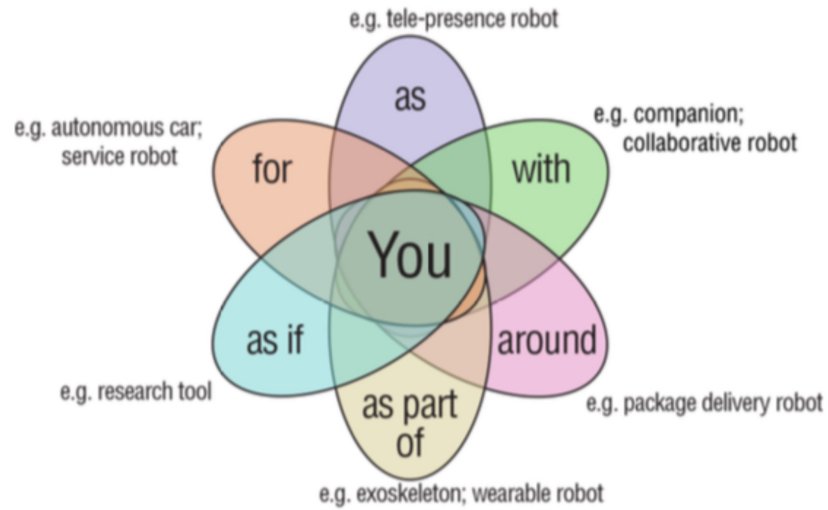
Intelligence

Ability to determine behavior that will maximize the likelihood of goal satisfaction under dynamic and uncertain conditions, linked to the environment and the interaction with other agents

Dimensions

1. Perception of environment and interaction partners
2. Modeling of environment and humans
3. Planning actions to interact with environment and humans
4. Executing plans under physical and social constraints
5. Learning through interaction with the environment or humans







1. Remote HRI
2. Co-located HRI
3. Physical HRI

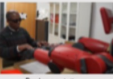
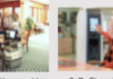
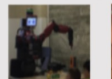
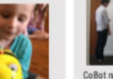
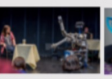
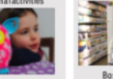
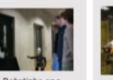


1. **Timespan**
 - a) Short-term
 - b) Medium-term
 - c) Long-term
 - d) Life-long
2. **Duration**
3. **Frequency**





Context

Healthcare and therapy				Industry	
					
Nao and child with ASD interacting [16]	Para emotionally assisting the elderly [168]	Baxter assisting a blind person [31]	Baxter being synthetically taught in a factory		
					
Robota assisting a child with ASD [29]	Pearl assisting an elder person [147]	SelfiQ assisting an elder person	Rob ear carrying a patient	Locusbots™ collaboratively operating in a warehouse	
Education, entertainment and art		Home and workplace		Search and rescue	
					
Baxter teaching children [67]	Bee-bot used for educational activities	Cobot navigating an office corridor [19]	Care-O-bot 4 in a home	Insuktan & Packbot equipped with social behavior [25]	
					
HERB acting in a play [209]	Furby with a child	Bossa Nova's supermarket robot	HERB engaging in kitchen tasks	Survivor buddy/Insuktan in a simulated disaster environment [181]	
Public service				Social sciences	
					
Robovie in a shopping mall [170]	Roboceptionist at department reception [79]	Pepper at a store entrance	Robotinho on a museum tour [63]	Cog used to study human cognition	Robo-la used to study child development [53]



Problem: Loneliness has a negative impact on physiological and mental health

Skills and purpose

- Evoke positive emotions
- Stimulating and motivating influence
- Assist people (with disabilities) at home
- Learn from social environment
- Autonomy and recovery

Challenges

- What are the actual user needs?
- Will people use the robot long-term?
- How can the robot socially evolve?



<https://nerdreactor.com/2019/01/14/best-companion-robots-of-ces-2019/>



https://globalsmtseasia.com/show_central/companion-robots-featured-at-shanghai-electronics-show/

Problem: Evaluation of social development is difficult in natural settings

Advantages

- Operationalizing theoretical assumptions
- Model structures can be manipulated and examined
- Experiments can be repeated with identical conditions
- Alternative hypotheses can easily be examined
- Hazardous conditions can be tested



<https://www.facebook.com/iCub-Humanoid/>

Limitations

Robots cannot fully represent humans

- Experimentation on limited model features (perception, attention, cognition)
- Some processes cannot be implemented (empathy, irrational decisions)



<https://www.itec.rwth-aachen.de/cms/ITEC/Forschung/~lrkjh/Teilnahme-an-Studien/lidx/1/>

50

Problem: Therapy of certain disorders needs to be provided at a high frequency and in very personalized ways

Skills and purpose

- Interactivity
- Stimulating and motivating influence
- Humane technological contribution
- Permanent social interaction possible
- Teaching social interaction skills

Disorders

- Dementia: Affiliation, permanent social contact
- Stroke / Rehabilitation: Sensorimotor exercises
- Autism: Learning social skills



Problem: Children with different learning abilities and speeds are in need of individualized support

Skills and Purpose

- Attract attention
- Playful engagement
- Non-threatening teaching experience
- Programmable → STEM education
- Standardized interaction possible

Limits

- Usability very limited and integration in school environment difficult
- Expensive
- Breaks easily



<https://bold.expert/the-social-robot-teacher/>

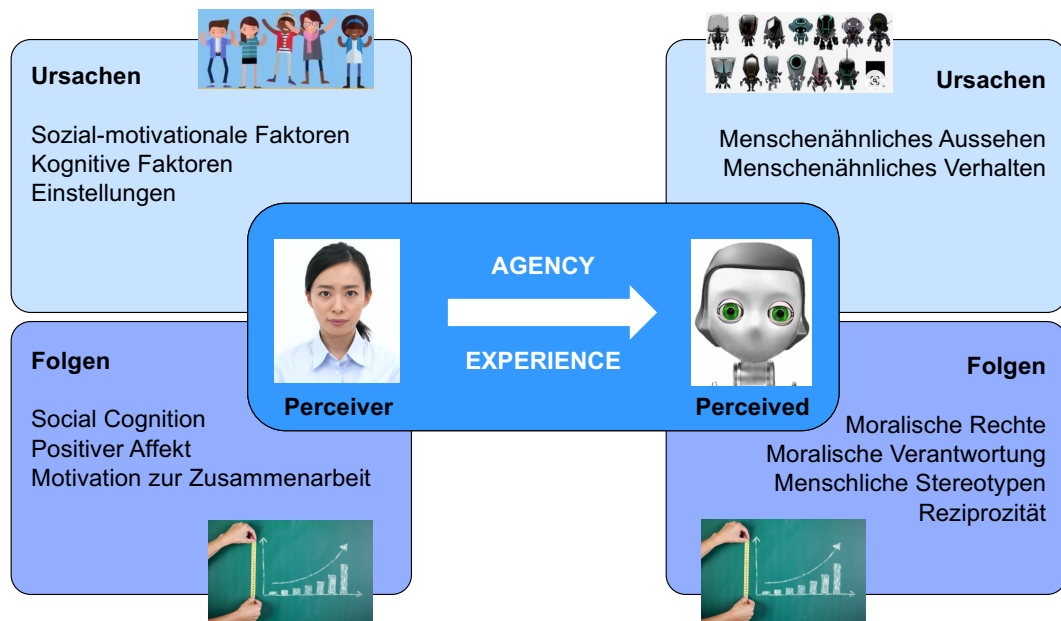


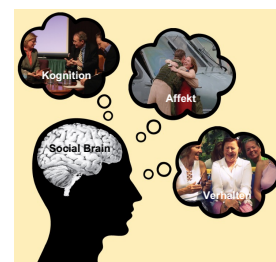
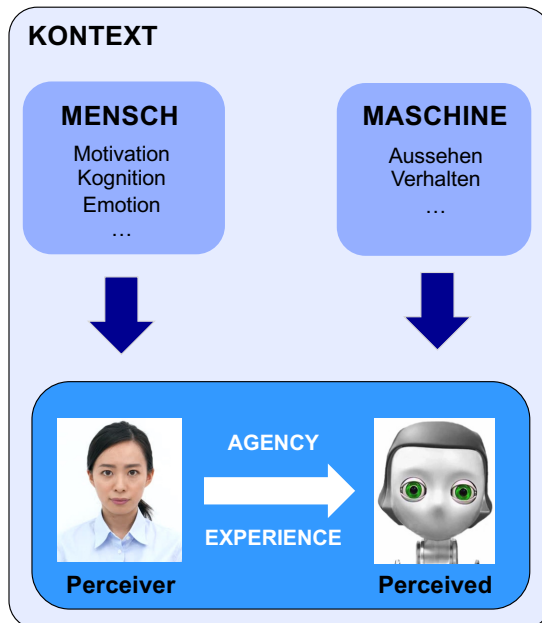
<https://www.timeslive.co.za/sunday-times/lifestyle/2017-09-18-robots-could-replace-teachers-within-the-next-decade/>

- Soziale Roboter werden am **Arbeitsplatz**, in der **Forschung** und im **täglichen Leben** eingesetzt. Je nach Einsatzort, muss das Design der Roboter angepasst werden
- Das Design des Roboters beeinflusst, wie er **wahrgenommen wird** und wie mit ihm **interagiert wird**. Vor allem schlecht kalibrierte Vorannahmen müssen durch gutes Design verbessert werden
- Bei Robotern kommt es nicht nur auf das **Design des Aussehens** an, sondern auch auf **Interaktionsparameter**. Letztere beeinflussen auch, wie stark sozial-kognitive Prozesse angesprochen werden können
- **Anwendungsbereiche** für soziale Robotik sind: Forschung, Lehre, Therapie und Assistenz

Teil 2: Soziale Kognitionsforschung mit Robotern





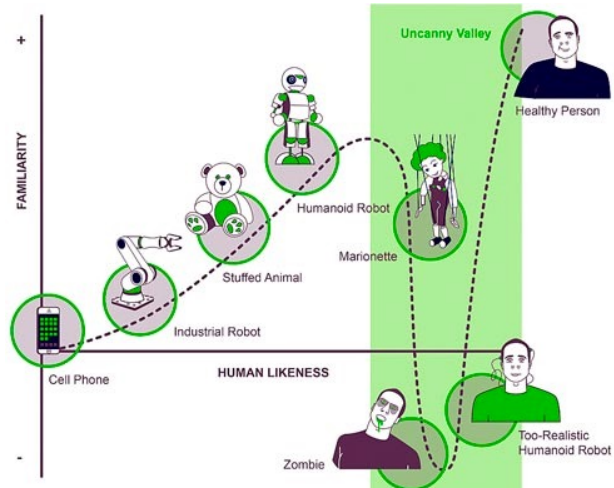




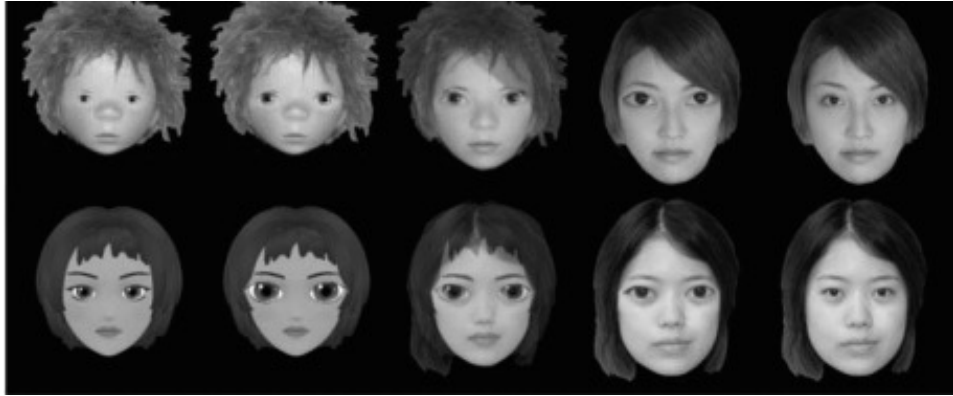
Beispiel 1: Anthropomorphismus



Human?

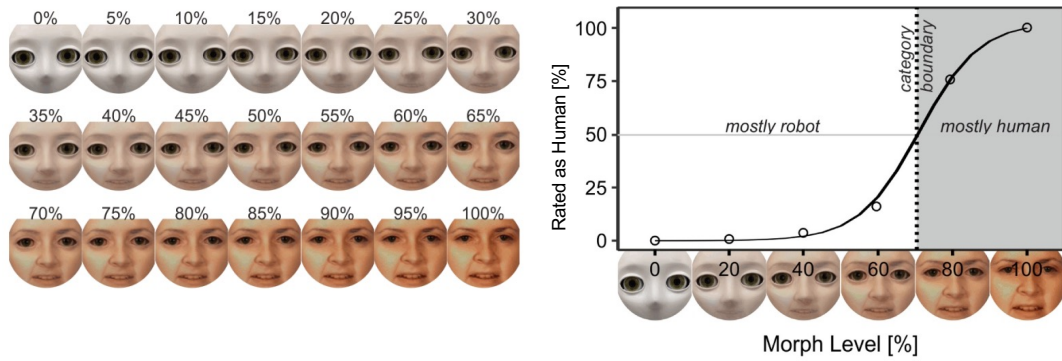


Kaetsyri et al. (2015)



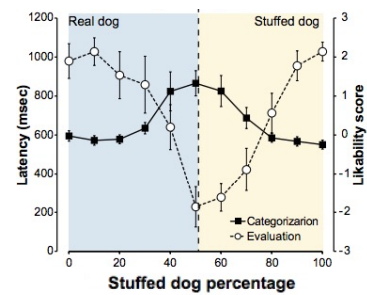
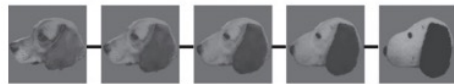
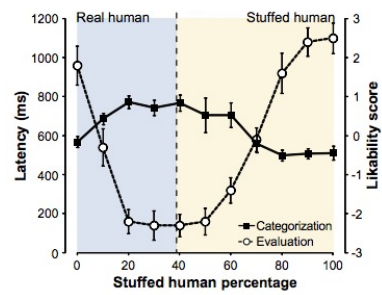
Perceptual Mismatch Hypothesis

Seyama & Nagayama (2007)

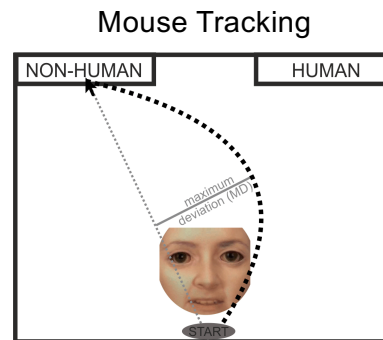
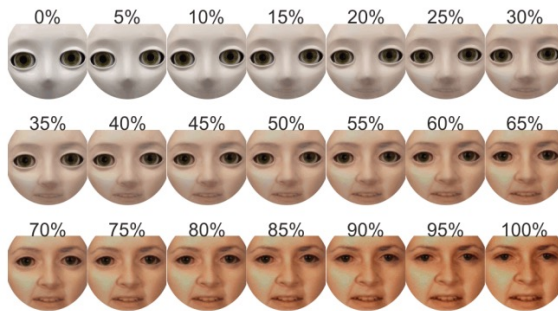


Categorical Perception Hypothesis

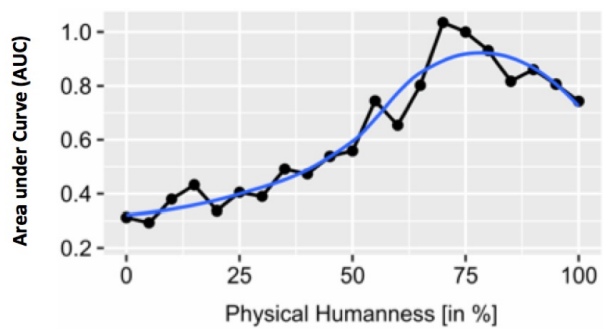
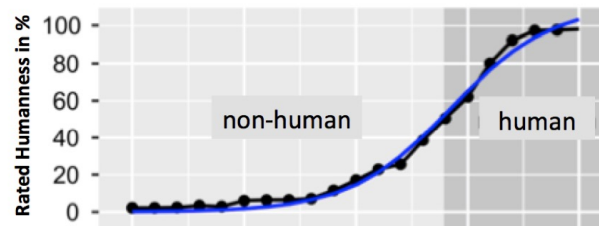
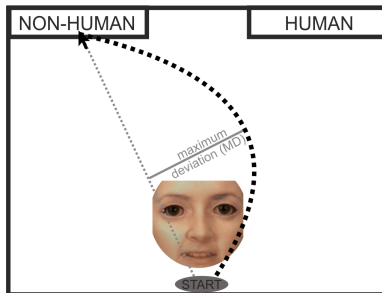
Looser & Wheatley (2010)



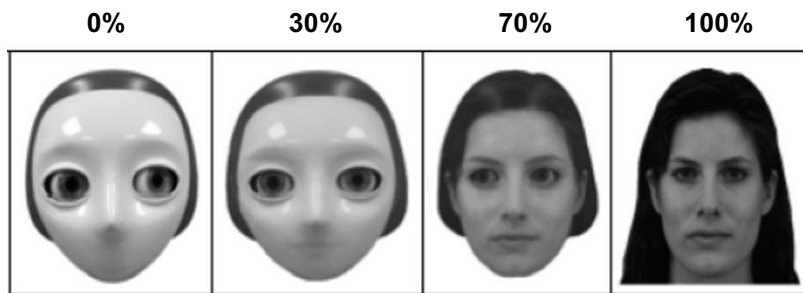
Yamada et al. (2013)



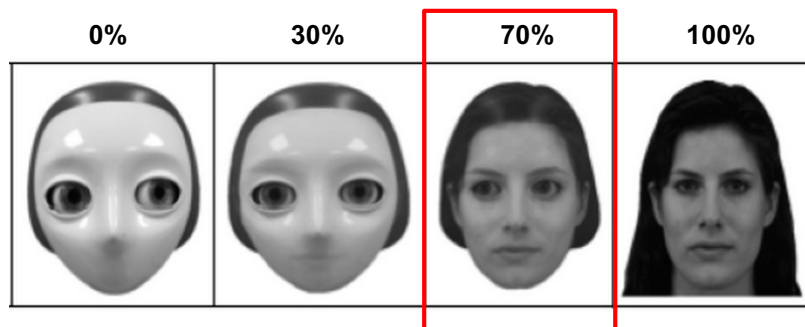
Mouse Tracking



Weis & Wiese (2017)

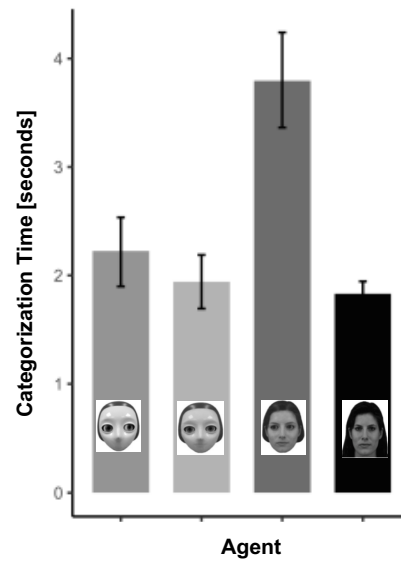
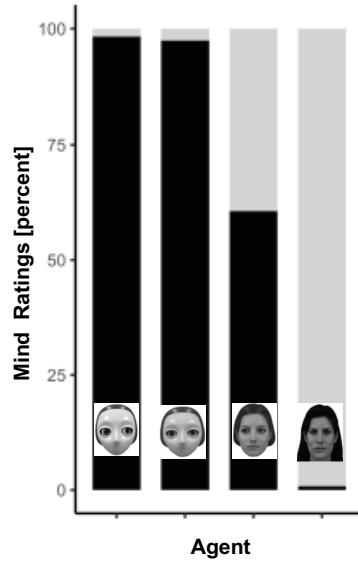


Wiese et al. (2018)

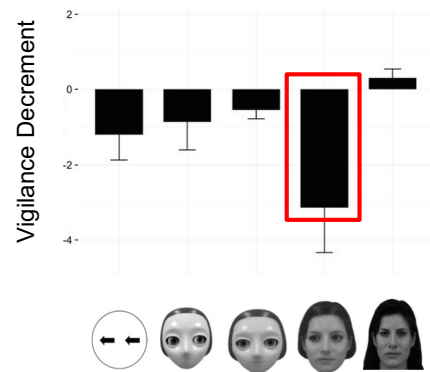
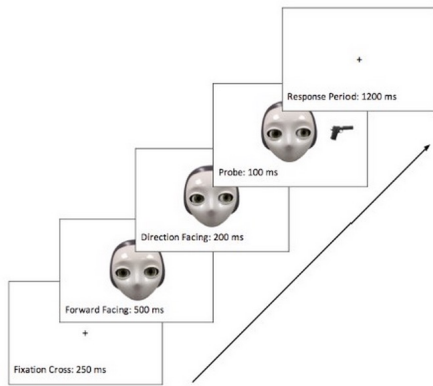


Kognitiver Konflikt?

Wiese et al. (2018)



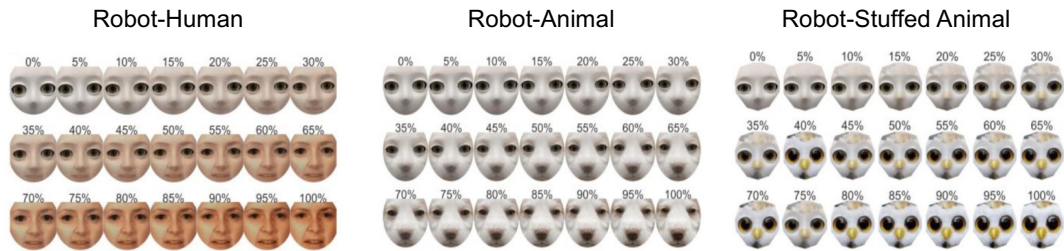
Wiese et al. (2018)



Wiese et al. (2018)



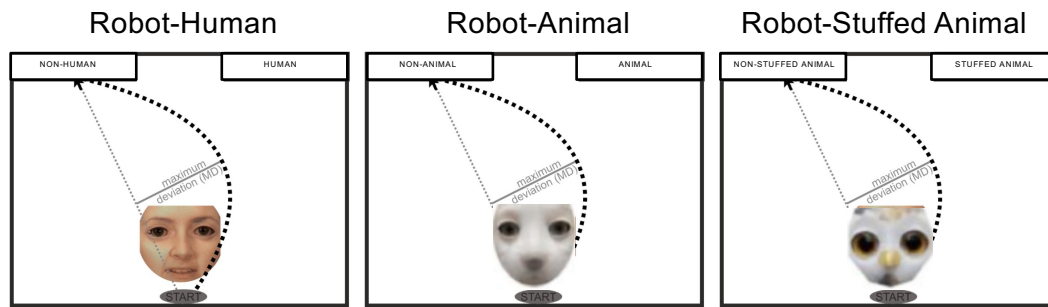
Beispiel 2: Ingroup-Outgroup



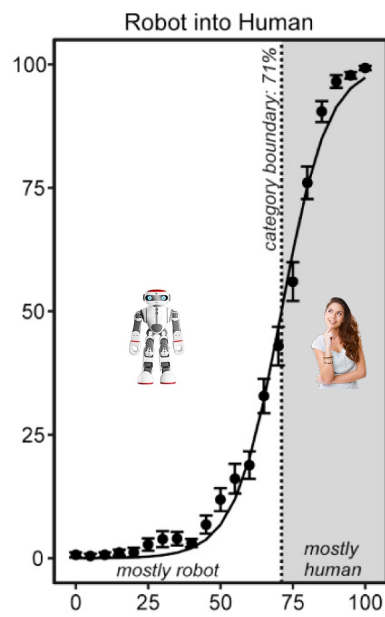
Wiese & Weis (2020)



Wiese & Weis (2020)

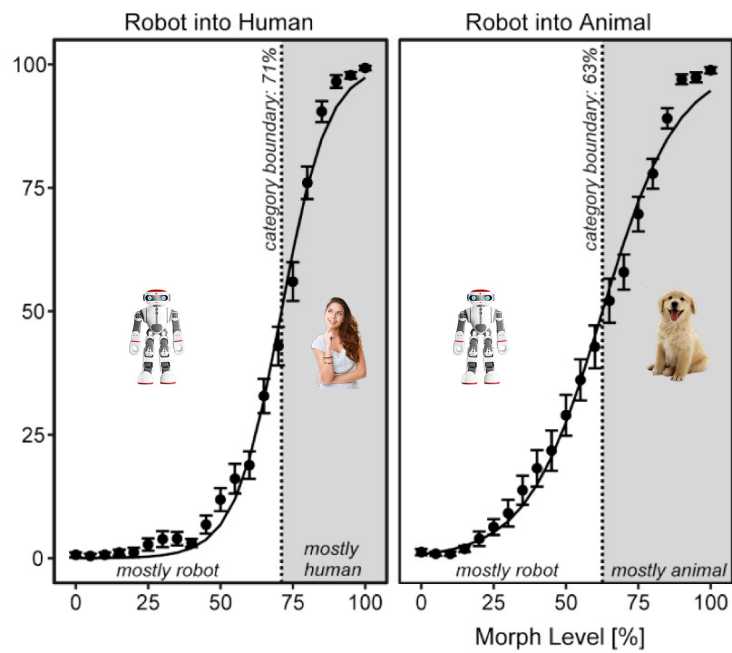


Wiese & Weis (2020)

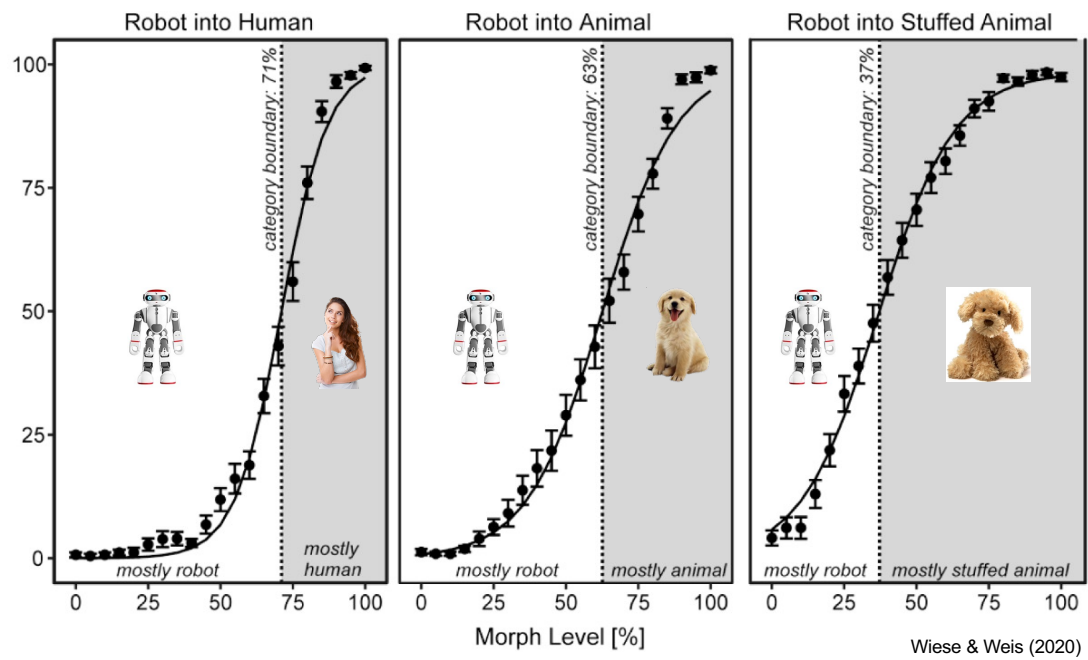


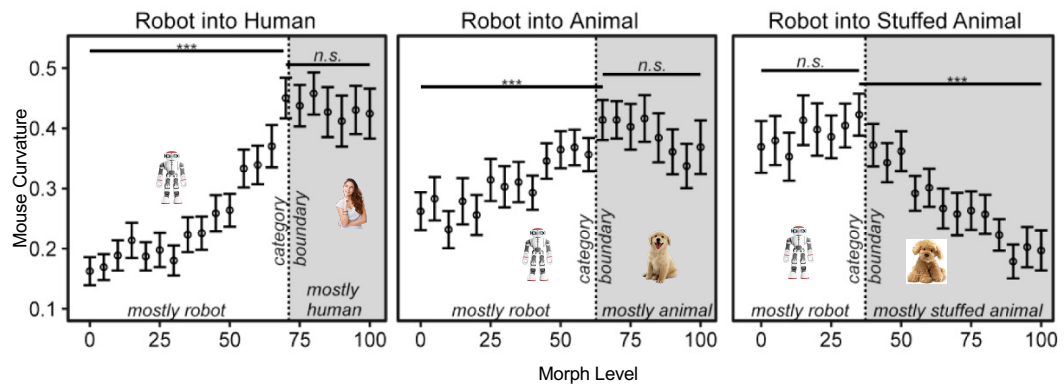
Morph Level [%]

Wiese & Weis (2020)

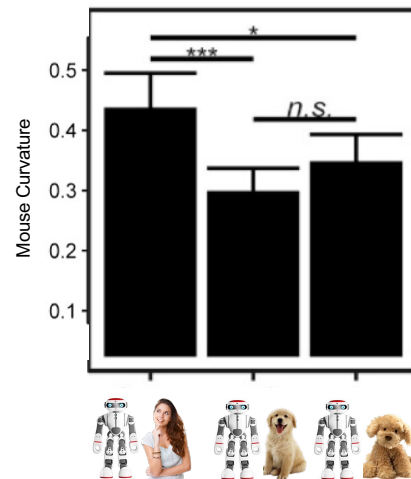


Wiese & Weis (2020)





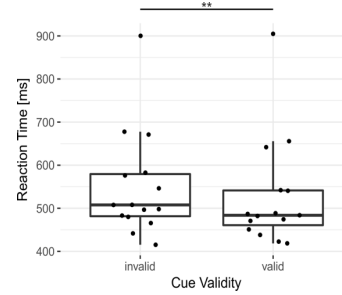
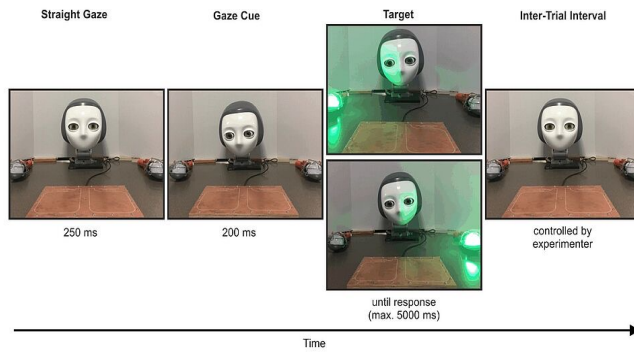
Wiese & Weis (2020)

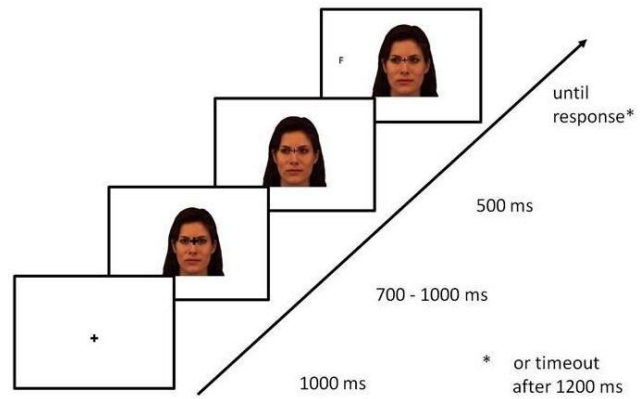
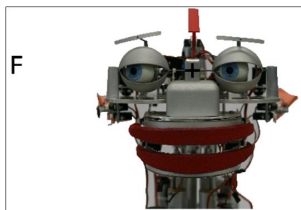
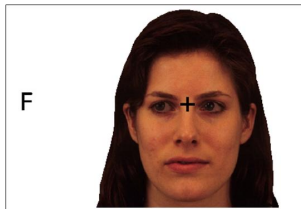


Wiese & Weis (2020)

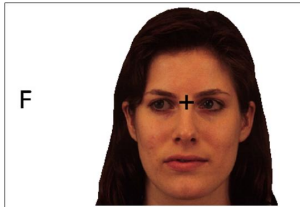


Beispiel 3: Gaze Following





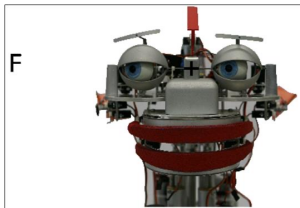
Wiese, Wykowska, Zwickel & Müller (2012)



Experimentelle Bedingungen

Gruppe 1

Human – Robot



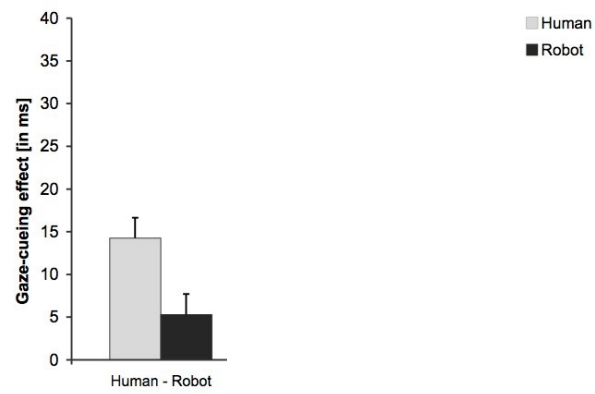
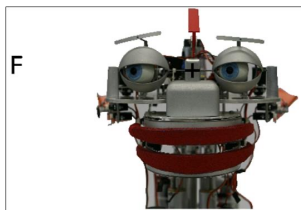
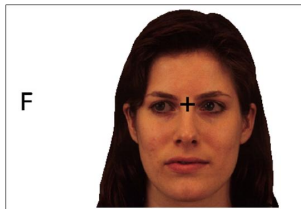
Gruppe 2

Human – Human-controlled Robot

Gruppe 3

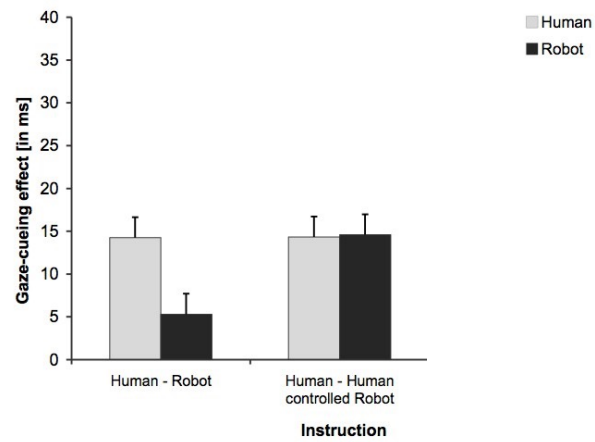
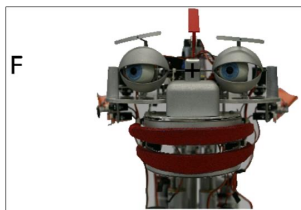
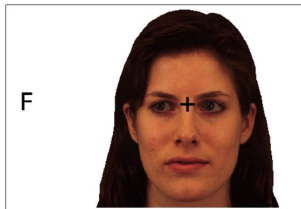
Mannequin - Robot

Wiese, Wykowska, Zwickel & Müller (2012)

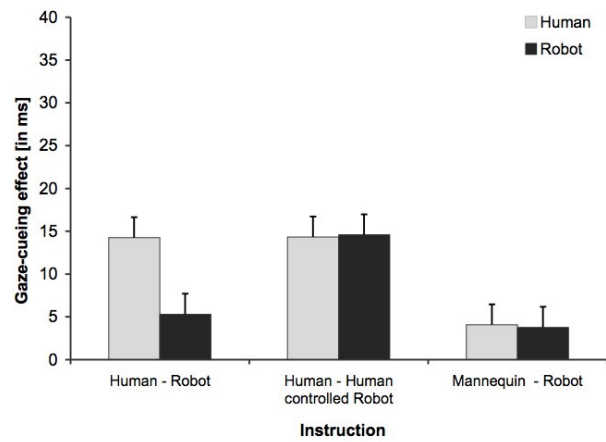
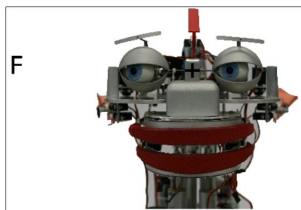
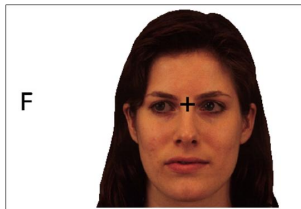


Instruction

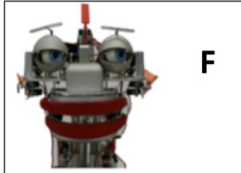
Wiese, Wykowska, Zwickel & Müller (2012)



Wiese, Wykowska, Zwickel & Müller (2012)

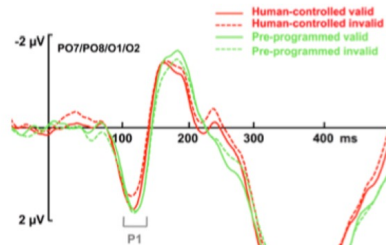
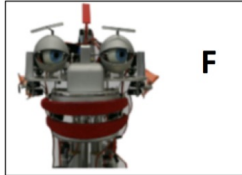


Wiese, Wykowska, Zwickel & Müller (2012)

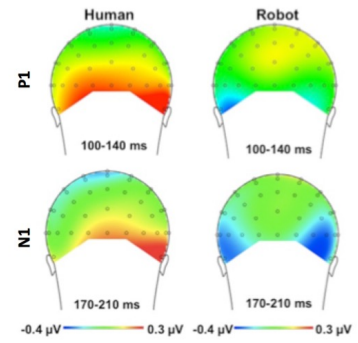
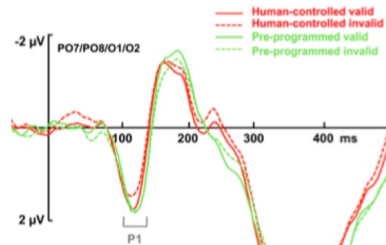
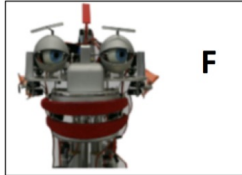


Human-controlled
Pre-programmed

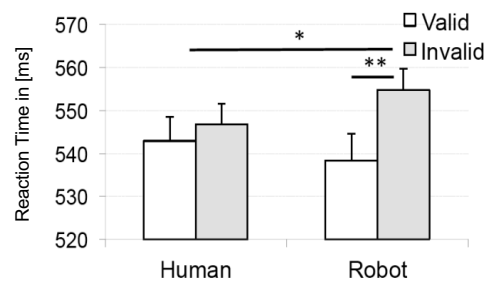
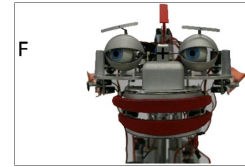
Wykowska, Wiese, Prosser & Mueller (2014)



Wykowska, Wiese, Prosser & Mueller (2014)



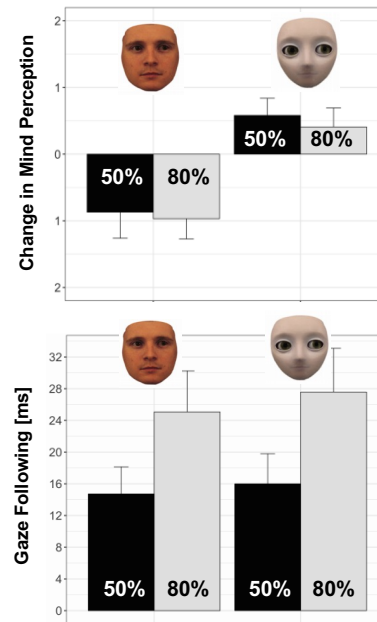
Wykowska, Wiese, Prosser & Mueller (2014)



Wiese, Wykowska & Mueller (2014)

Beispiel 3: Gaze Following

		Behavior	
		Reliable	Random
Appearance	Robot	 80% T	 50% T
	Human	 80% T	 50% T



Abubshait & Wiese (2017)

- Soziale Roboter werden in der Forschung eingesetzt um
 - soziale Interaktionen **unter kontrollierten Bedingungen** zu untersuchen
 - heraus zu finden, wie Roboterdesign **kognitive Parameter** beeinflusst
- Hierzu werden **klassische Paradigmen** aus der sozialen Kognitionsforschung eingesetzt und an die Mensch-Roboter Interaktion **angepasst**
- **Aussehen** und **Verhalten** des Roboters haben einen großen Einfluss darauf, wie er wahrgenommen wird und wie Menschen mit ihm interagieren
- Aussehen und Verhalten können Einstellungen und Interaktionsmuster **unterschiedlich beeinflussen**

- Soziale Roboter werden in vielen Bereichen bereits eingesetzt, was es notwendig macht, sich Gedanken über ein **geeignetes Design** und **geeignete Evaluationsmethoden** zu machen
- **Kein Design ist das Beste**, es kommt auf die Usergruppen, das Vorwissen und den Anwendungskontext an
- Roboter sind **hervorragende Plattformen**, um Mechanismen der sozialen Kognition zu untersuchen
- Unterschiedliche Designparameter beeinflussen bestimmte Parameter der sozialen Kognition unterschiedlich
 - Aussehen beeinflusst **Einstellungen**
 - Verhalten beeinflusst **Performanz**